Statistics 280 — Practice Midterm 2C

Unofficial practice exam in the style of the W25 and W26 Midterm 2.
Covers Lectures 9–15 (Chapters 13–17): probability, Bayes' rule, counting, binomial and hypergeometric distributions, random variables, expected value, standard error, covariance, box models, and the normal approximation, plus R for probability and simulation.

Name _____ Start time / End time _____

- Answer all questions in the space provided and indicate where you have written additional work.
- Solutions without justification will be marked as wrong for all questions except for questions asking for R code. Show all work. A calculator command does not count as justification.
- Question 7 is an extra credit question (15 points).
- Express answers as fractions or as decimals rounded to 2 decimal places.
- The last page is a table of values from the normal distribution. Closed book, no notes, calculator allowed.
- Suggested time: 80 minutes. Total: 200 points + 15 extra credit.

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**Total**
Points	30	30	30	40	30	40	15	200
Score								

1. (30 points) Four friends play Secret Santa. Their four names go into a hat, and each friend in turn draws one name at random without replacement. A friend who draws their own name is "stuck".

a) (4 points) What is the chance that the first friend is stuck?

b) (8 points) What is the chance that nobody is stuck? (Hint: count the orderings of the 4 names with no one in their own position.)

c) (6 points) What is the chance that exactly 2 friends are stuck? Exactly 3?

d) (6 points) Let S be the number of stuck friends. Find E(S) without finding the distribution of S. (Hint: write S as a sum of four 0-1 random variables.) Why does this work even though those random variables are dependent?

e) (6 points) Fill in the blanks so that the code estimates P(nobody is stuck) and E(S) from 100,000 simulations.

```
santa = function() {
  draw = sample(______)
  return(sum(______))
}
s = numeric(100000)
for (i in 1:100000) { s[i] = santa() }
p_none = ______
ES = ______
```

2. (30 points) A survey of 1,000 voters asks about three ballot measures. 55% support X, 40% support Y, and 30% support Z. 20% support both X and Y, 15% support both X and Z, 10% support both Y and Z, and 5% support all three.

a) (6 points) What percentage support at least one of the three measures?

b) (8 points) What percentage support exactly one measure? Exactly two?

c) (5 points) What is P(Z | X)? Are support for X and support for Z independent?

d) (5 points) Draw a Venn diagram for X, Y and Z with the number of voters in each of the eight regions.

e) (6 points) Suppose you knew only the three percentages 55%, 40% and 30%. What would be the maximum and minimum possible percentages supporting all three? Supporting at least one?

3. (30 points) A disease affects 1 in 200 people. A screening test has P(+ | disease) = 0.95 and P(+ | no disease) = 0.10. Repeated tests on the same person are independent given the person's disease status.

a) (5 points) What is the chance that a randomly chosen person tests positive?

b) (6 points) Given a positive test, what is the chance the person has the disease? Use the odds form of Bayes' rule.

c) (6 points) The person is tested again and is positive again. What is the chance they have the disease now?

d) (6 points) How many positive tests in a row would be needed for the chance of disease to exceed 90%?

e) (4 points) If instead the first test is positive and the second is negative, what is the chance of disease?

f) (3 points) Is the independence assumption realistic? Briefly explain.

4. (40 points) A shipment of 50 phones contains 6 defective phones. An inspector tests 8 phones chosen at random without replacement.

a) (6 points) What is the chance that the inspector finds no defective phones? Give the formula and an R command.

b) (4 points) Write an R command for the chance that the inspector finds at least 2 defective phones.

`R command:` _____

c) (4 points) The shipment is accepted if the inspector finds at most 1 defective phone. Write an R command for the chance it is accepted.

`R command:` _____

d) (6 points) At a different factory, 12% of phones are defective, independently. What is the chance that exactly 1 of 8 phones is defective? Give the formula, the value, and an R command.

e) (8 points) The factory in part d) ships 500 phones. Find the expected value and standard error of the number of defective phones, and use the normal approximation with the continuity correction to estimate the chance that between 50 and 70 phones, inclusive, are defective.

f) (6 points) Write R commands for the exact chance in part e) and for the normal approximation in part e).

`R command:` _____

`R command:` _____

g) (6 points) Review 15 gives a rule of thumb for when the normal approximation works for the number of 1's in n draws. Does part d) satisfy it? Does part e)? What goes wrong if you use the normal approximation for P(0 defective in 8)?

5. (30 points) You roll a die repeatedly until the running total of the spots reaches 10 or more. Let N be the number of rolls needed.

a) (10 points) Fill in the blanks so that the function simulates N.

```
rolls_needed = function(target) {
  total = ______
  n = ______
  while (______) {
    total = ______
    n = ______
  }
  return(______)
}
```

b) (6 points) Write R code that uses `rolls_needed` to estimate E(N), SE(N), and P(N ≤ 3) from 100,000 simulations.

c) (6 points) Find P(N = 1) and P(N = 2) exactly.

d) (8 points) Let M be the maximum of two dice. Find P(M = m) for m = 1, …, 6 (give a formula) and E(M).

6. (40 points) A box contains five tickets: 2, 4, 4, 6, 9.

a) (6 points) Find the average and SD of the box.

b) (8 points) You draw 25 tickets with replacement. Find the expected value and standard error of the sum, and of the average, of the draws.

c) (8 points) Estimate the chance that the average of the 25 draws is at least 5.5, using the normal approximation for the sum with the continuity correction.

d) (8 points) Now draw 3 tickets *without* replacement. What is the expected value of their sum? Can you use $\sqrt{n} \times$ SD of box for its standard error? Write R code that estimates the SE of the sum by simulation.

e) (6 points) How many draws with replacement would you need for the SE of the average to be at most 0.1?

f) (4 points) Compared with 25 draws, what happens to the SE of the sum and the SE of the average if you make 100 draws with replacement?

7. (15 points extra credit)

Using $\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$ and the rules for expected values, show that $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$. Then use this to show that if $\text{Var}(X) = \text{Var}(Y)$, then $\text{Cov}(X + Y, X - Y) = 0$.

A Normal Table

Height = height of the standard normal curve at z (percent per standard unit). Area = percentage of the area under the normal curve between −z and z.

z	Height	Area	z	Height	Area	z	Height	Area
0.00	39.89	0.00	**1.50**	12.95	86.64	**3.00**	0.443	99.730
0.05	39.84	3.99	**1.55**	12.00	87.89	**3.05**	0.381	99.771
0.10	39.70	7.97	**1.60**	11.09	89.04	**3.10**	0.327	99.806
0.15	39.45	11.92	**1.65**	10.23	90.11	**3.15**	0.279	99.837
0.20	39.10	15.85	**1.70**	9.40	91.09	**3.20**	0.238	99.863
0.25	38.67	19.74	**1.75**	8.63	91.99	**3.25**	0.203	99.885
0.30	38.14	23.58	**1.80**	7.90	92.81	**3.30**	0.172	99.903
0.35	37.52	27.37	**1.85**	7.21	93.57	**3.35**	0.146	99.919
0.40	36.83	31.08	**1.90**	6.56	94.26	**3.40**	0.123	99.933
0.45	36.05	34.73	**1.95**	5.96	94.88	**3.45**	0.104	99.944
0.50	35.21	38.29	**2.00**	5.40	95.45	**3.50**	0.087	99.953
0.55	34.29	41.77	**2.05**	4.88	95.96	**3.55**	0.073	99.961
0.60	33.32	45.15	**2.10**	4.40	96.43	**3.60**	0.061	99.968
0.65	32.30	48.43	**2.15**	3.96	96.84	**3.65**	0.051	99.974
0.70	31.23	51.61	**2.20**	3.55	97.22	**3.70**	0.042	99.978
0.75	30.11	54.67	**2.25**	3.17	97.56	**3.75**	0.035	99.982
0.80	28.97	57.63	**2.30**	2.83	97.86	**3.80**	0.029	99.986
0.85	27.80	60.47	**2.35**	2.52	98.12	**3.85**	0.024	99.988
0.90	26.61	63.19	**2.40**	2.24	98.36	**3.90**	0.020	99.990
0.95	25.41	65.79	**2.45**	1.98	98.57	**3.95**	0.016	99.992
1.00	24.20	68.27	**2.50**	1.75	98.76	**4.00**	0.013	99.9937
1.05	22.99	70.63	**2.55**	1.54	98.92	**4.05**	0.011	99.9949
1.10	21.79	72.87	**2.60**	1.36	99.07	**4.10**	0.009	99.9959
1.15	20.59	74.99	**2.65**	1.19	99.20	**4.15**	0.007	99.9967
1.20	19.42	76.99	**2.70**	1.04	99.31	**4.20**	0.006	99.9973
1.25	18.26	78.87	**2.75**	0.91	99.40	**4.25**	0.005	99.9979
1.30	17.14	80.64	**2.80**	0.79	99.49	**4.30**	0.004	99.9983
1.35	16.04	82.30	**2.85**	0.69	99.56	**4.35**	0.003	99.9986
1.40	14.97	83.85	**2.90**	0.60	99.63	**4.40**	0.002	99.9989
1.45	13.94	85.29	**2.95**	0.51	99.68	**4.45**	0.002	99.9991